Suppose that we are given a set S of truth assignments with an odd numbers of elements. Let D be the set of formulas of the language which a majority of truth assignments in S makes true. Which of the following is always true? Give reasons for each case.

(a) For an well-formed formula ϕ, either ϕ or $\neg\phi$ belongs in D.

This is true. Since if ϕ is not in D that means that less than half of the truth assignments of S make ϕ true. Therefore $\neg\phi$ results in more than half (since S has odd size) of the truth assignments giving $\neg\phi$ a true truth value and granting it entry to D.

(b) If ϕ belongs in D and $(\phi \to \theta)$ is a tautology, then θ belongs to D.

Suppose that more than half of the truth assignments in S make ϕ true, so that $\phi \in D$. Suppose that $(\phi \to \theta)$ is a tautology so all truth assignments in D make $(\phi \to \theta)$ true. Then the truth assignments of S which make ϕ truth must also make θ true granting it entry to D. So, (b) is true.

(c) If ϕ and $(\phi \to \theta)$, belong to D, then θ belongs to D.

Suppose ϕ and $(\phi \to \theta)$, belong to D. That means that the majority of truth assignments in S make ϕ and $(\phi \to \theta)$ true.

[scratch] suppose s has 9 elements

5 make ϕ true and 5 make $(\phi \to \theta)$ true But suppose that the 4 of the 5 truth assignments that make ϕ true make θ false So only in one truth assignment are both ϕ and θ true. In the remaining suppose both ϕ and θ are false.